

Lab- 2

Experiment Name: ALU circuit using simulator and HDL.

Objective:

- To understand the integration of arithmetic circuits (Adders/Subtractors) and logical gates.
- To implement a **Selection Control** mechanism (using Multiplexers) to switch between different operations.

Theory: The ALU can be thought of as the core processing unit of a computer, responsible for performing both arithmetic and logical tasks. It is essentially a combinational circuit made up of two main sections: an Arithmetic Unit and a Logic Unit. The Arithmetic Unit typically relies on full adders to handle operations like addition and subtraction, while the Logic Unit performs bitwise functions such as AND, OR, and XOR.

Both units operate on the same input data at the same time, but only one result is selected as the final output. This selection is done using a multiplexer (MUX), which is controlled by specific selection lines. These control signals act like operation codes, directing the MUX to choose the desired result and send it to the output while disregarding the others.

Circuit Diagram:

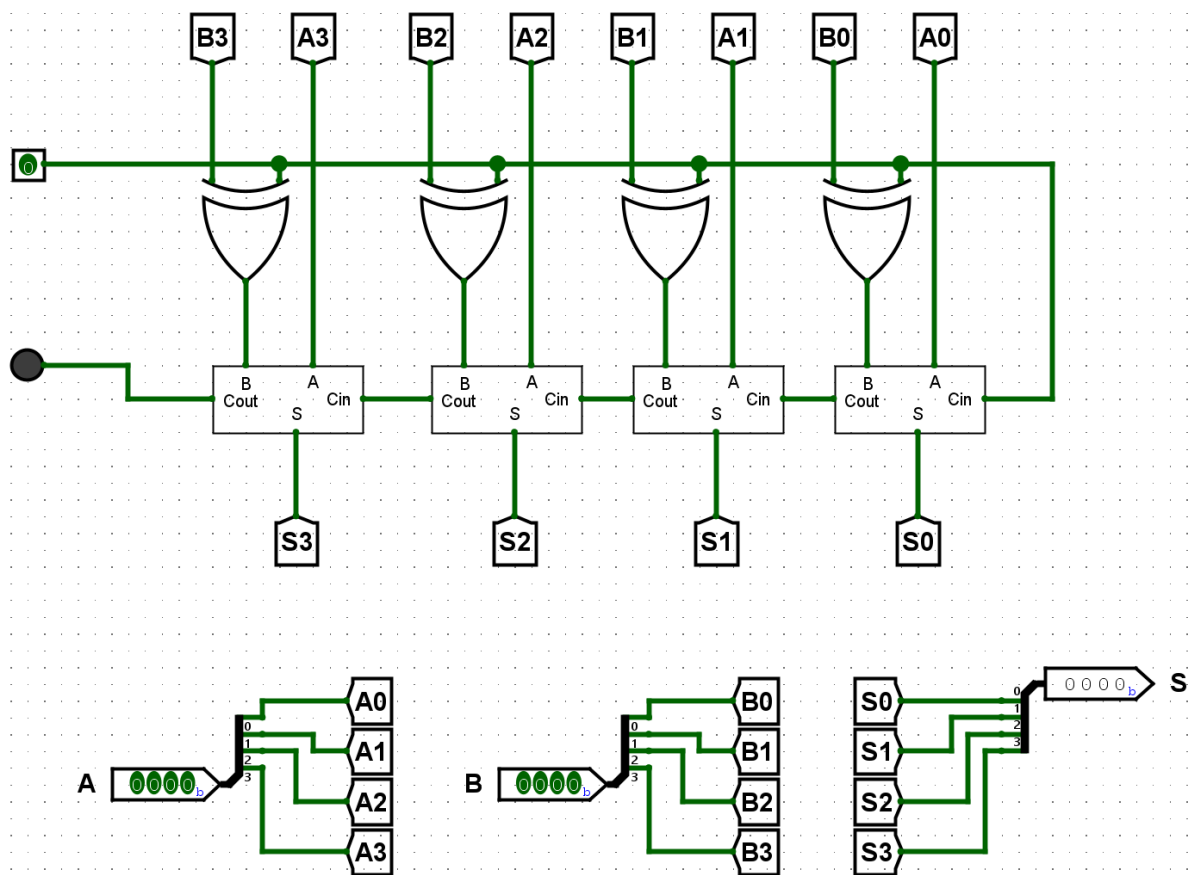


Fig 01: 4 Bit Adder Subtractor

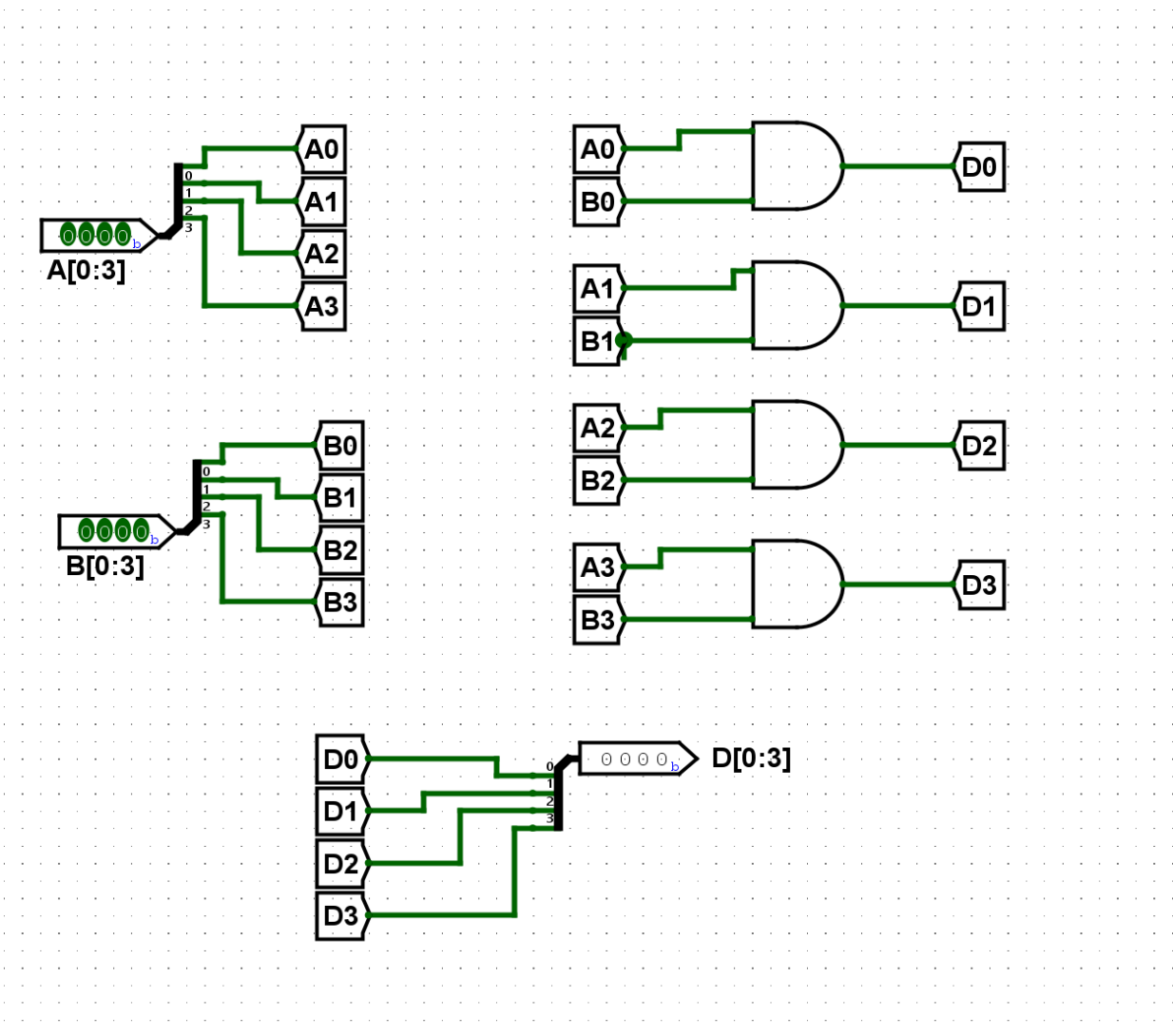


Fig 02: 4 Bit And Gate

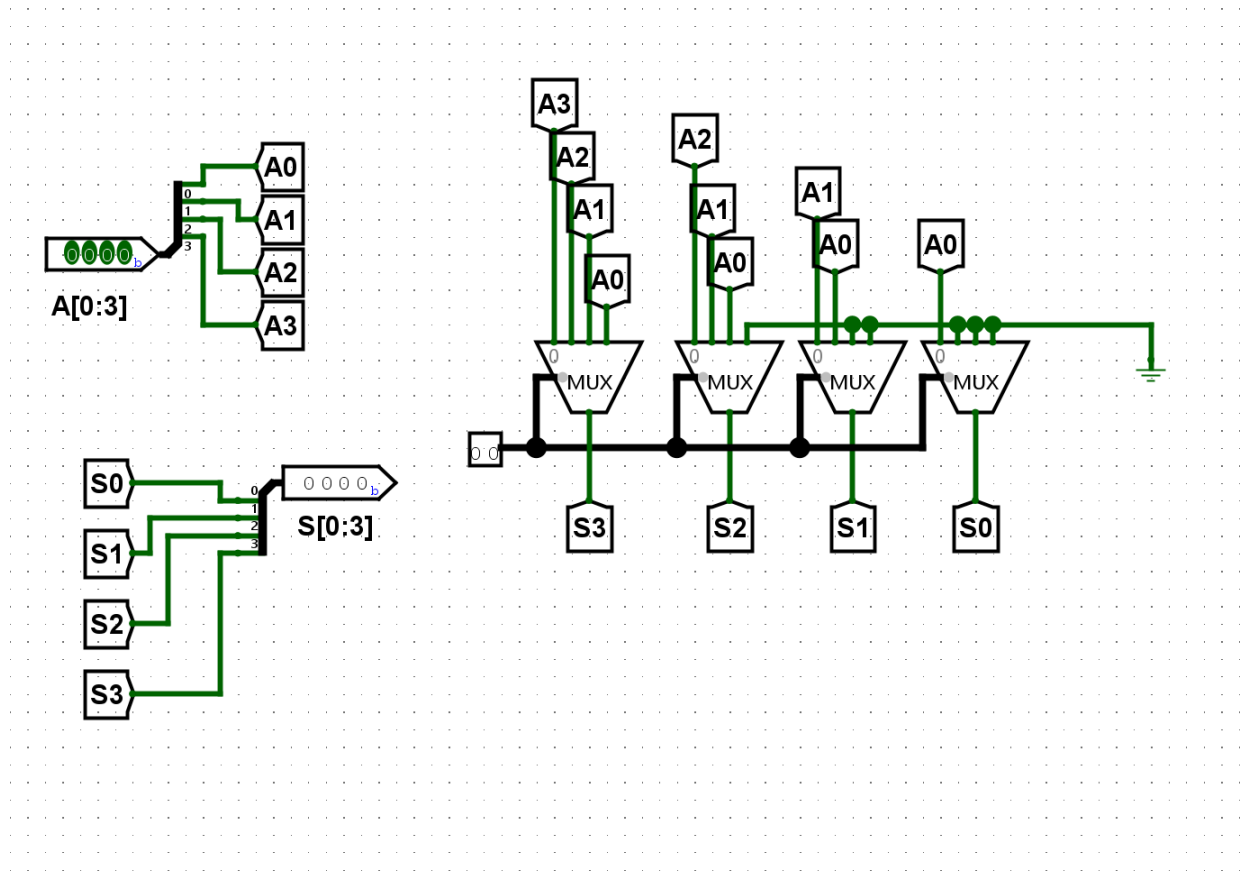


Fig 03: 4 Bit Left Shifter

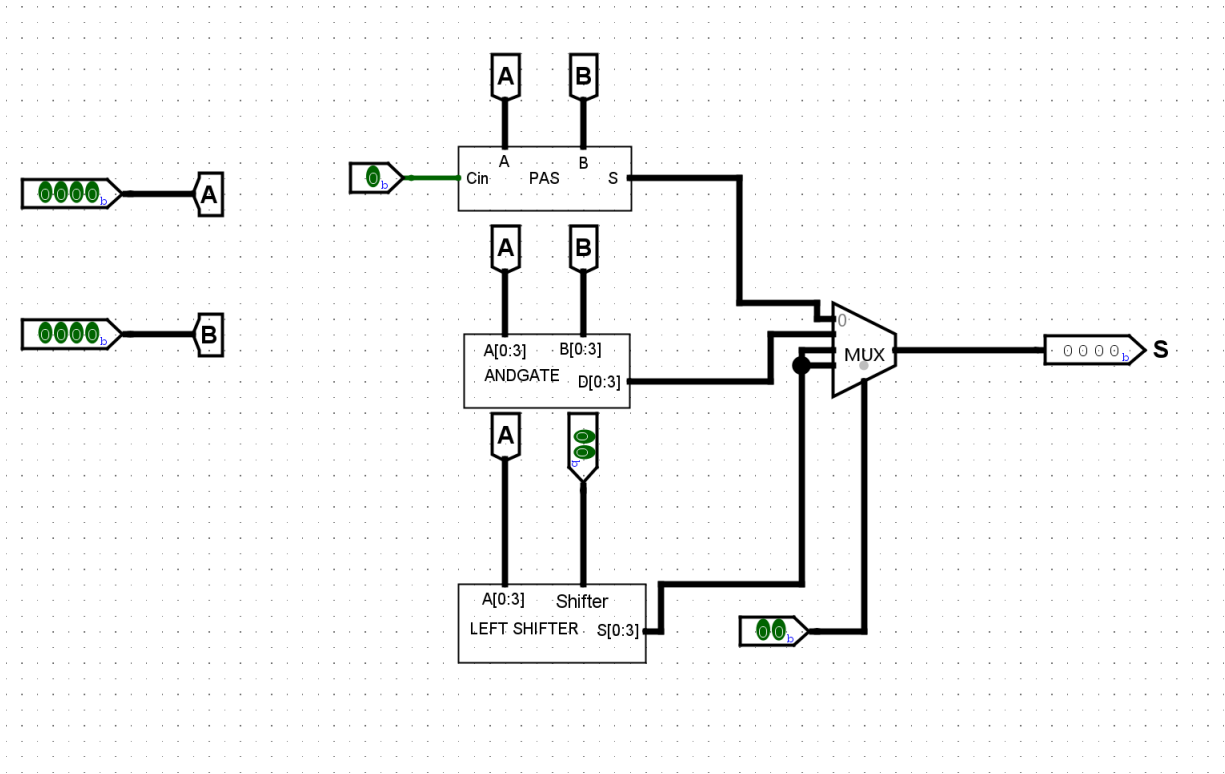


Fig 04: 4 Bit Arithmetic Logic Unit

Procedure:

- First, open your digital simulation software and place all the required components on the workspace. These include a 4-bit parallel adder-subtractor, four AND gates, a 4-bit left shifter, and a 4-to-1 multiplexer (MUX).
- Next, create two sets of 4-bit inputs named A (A3 to A0) and B (B3 to B0), which will be used for different operations.
- For arithmetic operations, connect inputs A and B to the adder-subtractor. Use a toggle switch as a mode control—set it to 0 for addition and 1 for subtraction.
- For logical operations, connect each corresponding bit of A and B to the four AND gates. This will allow you to perform a bitwise AND operation.
- Then, connect input A to the 4-bit left shifter to perform the shift operation.
- After that, connect the outputs of the adder-subtractor, AND gates, and shifter to a 4-to-1 multiplexer. Use the selection lines (S1 and S0) to choose which operation result will be displayed on the output.
- Finally, verify the circuit by entering sample binary values, such as 0011 and 0001. Change the selection lines and mode switch to check whether the results for addition, subtraction, AND, and left shift match the expected theoretical values.

Precautions:

- **Pin Mapping:** Ensure that the Most Significant Bit (MSB) and Least Significant Bit (LSB) are correctly mapped from the inputs to the components to avoid inverted results.
- **Mode Bit Conflict:** In the Adder-Subtractor, double-check that the Carry-in (C_{in}) is correctly synced with the control signal for subtraction (usually requires $C_{in} = 1$ for 2's complement).

Conclusion:

This experiment successfully demonstrated the design and working of a 4-bit ALU using three main functional units. These include an arithmetic unit (adder-subtractor), a logic unit (AND gates), and a shift unit (left shifter).

By integrating these components together, we were able to understand how different types of operations—such as arithmetic calculations, logical processing, and bit shifting—are performed in a single system. This experiment helped in visualizing how a processor handles multiple tasks using a unified ALU structure.